Molecules containing an adamantane moiety have a wide variety of applications, including drug delivery, transition metal catalysis, and supramolecular chemistry. Key steps in the synthesis of an adamantane-functionalized phthalimide scaffold are shown in Figure 1.

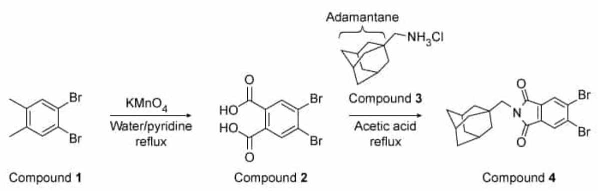


Figure 1 Synthesis of an adamantane-functionalized phthalimide scaffold

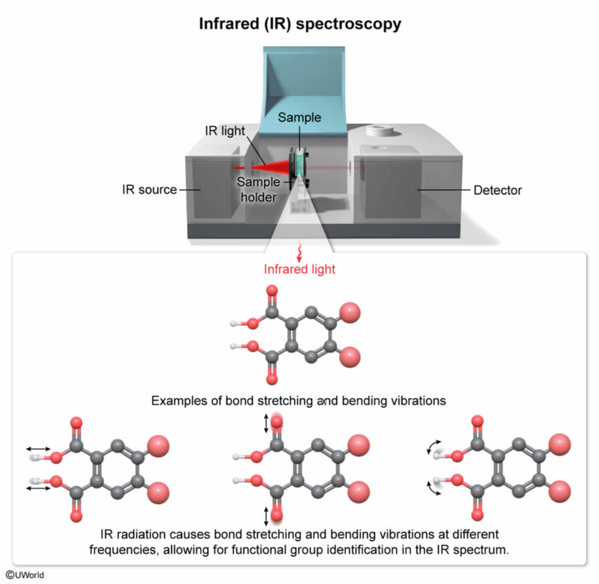
In the synthesis shown, Compound **1** was oxidized with $KMnO_4$ to form a dicarboxylic acid (Compound **2**). The structure of Compound **2** was confirmed by infrared (IR) spectroscopy rather than ^1H NMR spectroscopy because Compound **2** has poor solubility in organic solvent. The adamantane substituent was added to Compound **2** by reaction with Compound **3** under acidic conditions. The resulting product (Compound **4**) was analyzed by ^1H NMR spectroscopy, IR spectroscopy, and x-ray diffraction crystallography. The two bromides on the aromatic ring provide an avenue for further modification of the *N*-substituted phthalimide scaffold, potentially leading to the synthesis of new drugs.

Which of the following statements best explains why IR spectroscopy could be used to confirm the structure of Compound **2**?

- A. Different types of bonds absorb IR radiation at different frequencies, causing electronic excitations that distinguish the different functional groups in the compound. [19%]
- B. Different types of bonds absorb IR radiation at different frequencies, causing molecular fragmentation patterns that identify the different functional groups in the compound. [11%]
- C. The absorption of IR radiation by molecules causes molecular rotations at different frequencies, allowing for functional group identification. [4%]
- D. The absorption of IR radiation by molecules causes intramolecular stretching and bending vibrations of different bond types at different frequencies, allowing for functional group identification. [64%]

Correct
64% answered correctly

Explanation:



Infrared (IR) spectroscopy is a technique used to **identify functional groups** in a compound through the analysis of the absorption of IR radiation by **different types of bonds** present in the molecule. A sample is irradiated with IR light, and the spectrometer detects and records the amount of light that passes through the sample over a range of frequencies. The data is represented on the IR spectrum as the percent transmittance (*y*-axis) versus wavenumber (*x*-axis).

The IR radiation interacts with the different types of bonds within the molecule, causing stretching vibrations and rotations of the bonds. Functional groups absorb different amounts of IR light (ie, have different absorption intensities at different frequencies). As a result, IR spectroscopy can be used to confirm the formation of a reaction product by identifying the peaks unique to the newly formed functional group(s).

Therefore, the structure of Compound **2** can be confirmed using IR spectroscopy because the absorption of IR radiation by molecules causes **intramolecular** bending and stretching **vibrations** of different bond types at different frequencies, allowing for functional group identification.

(Choice A) Electronic excitation occurs during ultraviolet (UV) or visible light absorption rather than IR light absorption.

(Choice B) Molecular fragmentation occurs in mass spectrometry rather than IR spectroscopy.

(Choice C) The absorption of IR radiation by molecules causes stretching and bending *vibrations* rather than rotations of the molecules in the sample. Rotational motion results from interactions with *microwave radiation*.

Educational objective:
Infrared (IR) spectroscopy is a technique used to identify functional groups in a molecule through analysis of the absorption of IR radiation by different types of bonds

present. IR radiation causes intramolecular bending and stretching vibrations of different bond types at different frequencies and intensities.

Time spent:0 sec

QID:403492

Subject:
Organic Chemistry

System:
Separation Techniques, Spectroscopy, and Analytical Methods

Molecules containing an adamantane moiety have a wide variety of applications, including drug delivery, transition metal catalysis, and supramolecular chemistry. Key steps in the synthesis of an adamantane-functionalized phthalimide scaffold are shown in Figure 1.

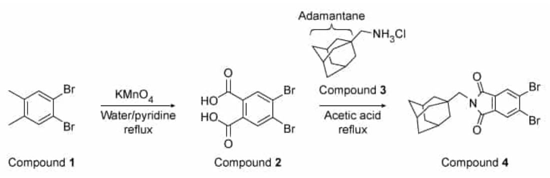
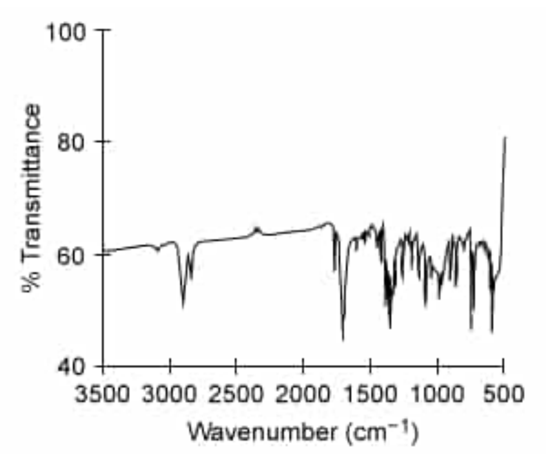


Figure 1 Synthesis of an adamantane-functionalized phthalimide scaffold

In the synthesis shown, Compound **1** was oxidized with $KMnO_4$ to form a dicarboxylic acid (Compound **2**). The structure of Compound **2** was confirmed by infrared (IR) spectroscopy rather than ^{1}H NMR spectroscopy because Compound **2** has poor solubility in organic solvent. The adamantane substituent was added to Compound **2** by reaction with Compound **3** under acidic conditions. The resulting product (Compound **4**) was analyzed by ^{1}H NMR spectroscopy, IR spectroscopy, and x-ray diffraction crystallography. The two bromides on the aromatic ring provide an avenue for further modification of the *N*-substituted phthalimide scaffold, potentially leading to the synthesis of new drugs.

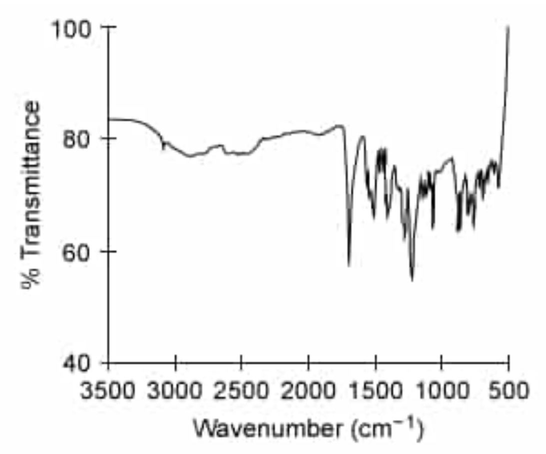
Which of the following spectra best illustrates the IR spectrum of Compound **2**?

A.



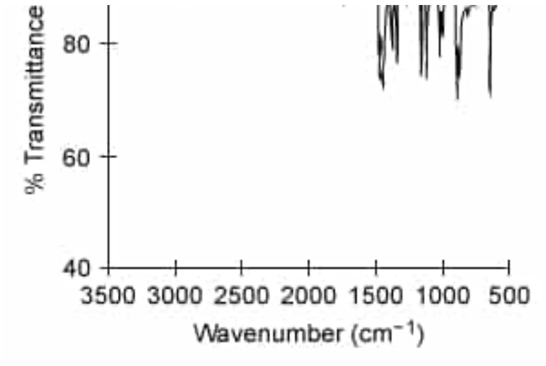
[39%]

✔ B.



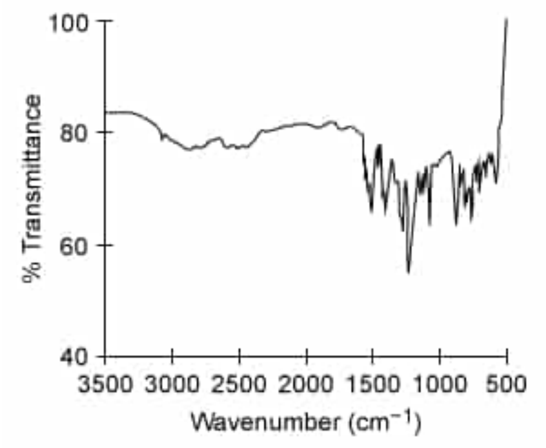
[45%]

C.



[6%]

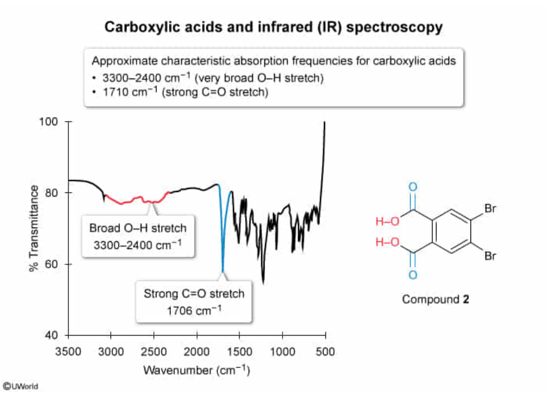
◯ D.



[8%]

Correct
46% answered correctly

Explanation:



In infrared (IR) spectroscopy, a sample is irradiated with IR light and a spectrometer detects how much light passes through the sample (ie, transmittance) over a range of frequencies. The data is recorded and displayed on an IR spectrum as the percent transmittance (*y*-axis) versus wavenumber (*x*-axis). Depending on the **types of bonds** present in a functional group, different amounts of IR light are absorbed (ie, the peak intensity varies) at different frequencies.

Carboxylic acids have two notable **characteristic absorptions** at approximately:

- **3300–2400 cm^{-1}** (very broad O–H stretch)
- **1710 cm^{-1}** (strong C=O stretch)

As seen in the structure in Figure 1, Compound **2** has two carboxylic acid substituents.

Therefore, the IR spectrum for Compound **2** must contain peaks corresponding to the carboxylic acid C=O and O–H stretches. The spectrum shown in Choice B has a very broad, irregular band between 3300–2400 cm^{-1} (corresponding to an O–H stretch) and a strong peak at 1706 cm^{-1} (corresponding to a C=O stretch).

(Choice A) This is the IR spectrum of Compound **4**. Although it contains a peak for a C=O stretch, it does not contain a broad band between 3300–2400 cm^{-1} corresponding to the carboxylic acid O–H stretch. Peaks of medium intensity near 2900 cm^{-1} and 2800 cm^{-1} correspond to the sp^3 C–H stretching from the adamantane substituent.

(Choice C) This spectrum does not contain either of the characteristic carboxylic acid absorptions (C=O stretch or O–H stretch), and instead is the IR spectrum of Compound **1**.

(Choice D) This spectrum is missing the strong peak at 1706 cm^{-1} corresponding to the carboxylic acid C=O stretch.

Educational objective:
Infrared (IR) spectroscopy measures the amount of light transmitted through a sample over a range of frequencies. Absorption frequencies and peak intensities depend on the type of bond present in a particular functional group. Carboxylic acids have two characteristic absorptions: 3300–2400 cm^{-1} (O–H stretch, broad and irregular band) and 1710 cm^{-1} (C=O stretch, strong peak).

Time spent:0 sec
Subject:
Organic Chemistry

QID:403493
System:
Separation Techniques, Spectroscopy, and Analytical Methods

Molecules containing an adamantane moiety have a wide variety of applications, including drug delivery, transition metal catalysis, and supramolecular chemistry. Key steps in the synthesis of an adamantane-functionalized phthalimide scaffold are shown in Figure 1.

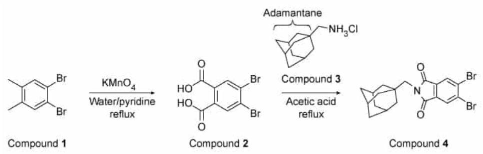
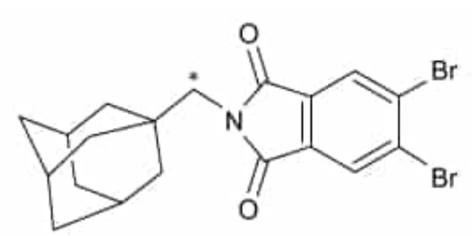


Figure 1 Synthesis of an adamantane-functionalized phthalimide scaffold

In the synthesis shown, Compound **1** was oxidized with $KMnO_4$ to form a dicarboxylic acid (Compound **2**).

The structure of Compound **2** was confirmed by infrared (IR) spectroscopy rather than ^1H NMR spectroscopy because Compound **2** has poor solubility in organic solvent. The adamantane substituent was added to Compound **2** by reaction with Compound **3** under acidic conditions. The resulting product (Compound **4**) was analyzed by ^1H NMR spectroscopy, IR spectroscopy, and x-ray diffraction crystallography. The two bromides on the aromatic ring provide an avenue for further modification of the *N*-substituted phthalimide scaffold, potentially leading to the synthesis of new drugs.



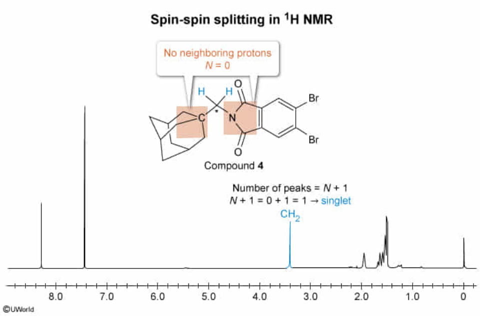
Compound **4**

In the ^1H NMR spectrum of Compound **4**, the protons bonded to the carbon atom marked with an asterisk appear as a:

- ✔ A. singlet. [54%]
- B. doublet. [33%]
- C. triplet. [11%]
- D. quartet. [1%]

Correct
53% answered correctly

Explanation:



微信Mai_ange

Proton nuclear magnetic resonance spectroscopy (^1H NMR) detects hydrogen atoms (protons) in a molecule by applying an external magnetic field and radio frequencies to a sample. Protons that exist in different magnetic environments are referred to as chemically **nonequivalent protons**. Interactions between neighboring, nonequivalent protons result in the **coupling** of these protons, which causes **spin-spin splitting** of signals in the ^1H NMR spectrum.

The splitting pattern of a signal (ie, the number of peaks into which a signal is split) is determined by applying the ***N* + 1** rule, where *N* is the **number of nonequivalent neighboring protons**. The area under the peak is proportional to the number of protons the peak represents. For example, if a methyl group ($-CH_3$) has one neighboring proton (ie, $N = 1$ and $N + 1 = 2$), the signal is split into two peaks (a doublet) and the area under the doublet is proportional to the three protons in the $-CH_3$ group.

Notably, coupling of protons on amines or alcohols with protons on neighboring carbons is usually negligible because amine and alcohol protons are exchangeable (ie, are transferred rapidly, not staying in place long enough to interact with neighboring protons).

The carbon atom in Compound **4** marked by an asterisk is a $-CH_2-$ group bonded to a quaternary carbon atom (ie, a carbon atom bonded to four other carbon atoms) and a tertiary amine. As such, the $-CH_2-$ group has no neighboring hydrogen atoms (ie, $N = 0$), and $N + 1 = 1$. Therefore, the $-CH_2-$ group is not coupled to any adjacent protons and appears as a singlet in the ^1H NMR spectrum.

(Choices B, C, and D) A doublet, triplet, or quartet arise when there is one, two, or three neighboring protons, respectively (ie, $N = 1$, 2, or 3, respectively).

Educational objective:
Interactions between neighboring, nonequivalent protons result in coupling of these protons, which causes spin-spin splitting of the corresponding signals in the ^1H NMR spectrum. The splitting pattern of a signal is determined by applying the $N + 1$ rule, where N is the number of neighboring nonequivalent protons.

Time spent:0 sec
Subject:
Organic Chemistry

QID:403494
System:
Separation Techniques, Spectroscopy, and Analytical Methods

Molecules containing an adamantane moiety have a wide variety of applications, including drug delivery, transition metal catalysis, and supramolecular chemistry. Key steps in the synthesis of an adamantane-functionalized phthalimide scaffold are shown in Figure 1.

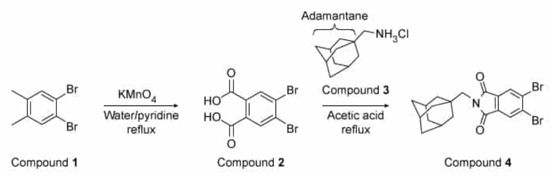


Figure 1 Synthesis of an adamantane-functionalized phthalimide scaffold

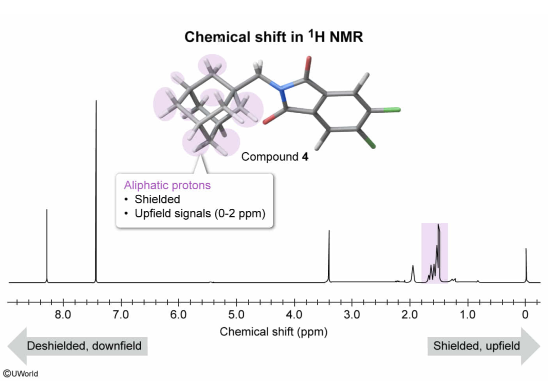
In the synthesis shown, Compound **1** was oxidized with $KMnO_4$ to form a dicarboxylic acid (Compound **2**). The structure of Compound **2** was confirmed by infrared (IR) spectroscopy rather than ^{1}H NMR spectroscopy because Compound **2** has poor solubility in organic solvent. The adamantane substituent was added to Compound **2** by reaction with Compound **3** under acidic conditions. The resulting product (Compound **4**) was analyzed by ^{1}H NMR spectroscopy, IR spectroscopy, and x-ray diffraction crystallography. The two bromides on the aromatic ring provide an avenue for further modification of the *N*-substituted phthalimide scaffold, potentially leading to the synthesis of new drugs.

Which range corresponds to the expected chemical shifts of the adamantane protons in the ^{1}H NMR spectrum of Compound **4**?

- A. 1.0–2.0 ppm [34%]
- B. 2.5–4.5 ppm [30%]
- C. 4.5–6.5 ppm [23%]
- D. 7.0–8.0 ppm [11%]

Correct
35% answered correctly

Explanation:



In proton nuclear magnetic resonance spectroscopy (^{1}H NMR), the chemical shift (ie, the location of the signals in the spectrum relative to a reference compound) is given in parts per million (ppm) with a typical range of 0–12 ppm. Tetramethylsilane (TMS) (ie, $Si(CH_3)_4$) is commonly used as a reference compound (ie, its peak is set to 0 ppm) in ^{1}H NMR because its methyl groups are very shielded, resulting in a signal that is upfield (ie, has smaller chemical shift) relative to most NMR signals.

The **chemical shift** of a proton depends on its **electronic environment**. An **upfield signal** arises when a proton is surrounded by an electron cloud, **shielding** it from the external magnetic field B_0. A **downfield** signal (ie, a larger chemical shift) arises when an adjacent electronegative substituent withdraws electrons from a proton, **deshielding** it from B_0.

The passage and the structure of Compound **4** in Figure 1 indicate that the adamantane substituent is a hydrocarbon (ie, made of carbon and hydrogen atoms). Aliphatic protons are surrounded by an electron cloud and shielded from B_0, resulting in an upfield chemical shift typically below 2 ppm. Therefore, the signals for the protons in the adamantane substituent in Compound **4** appear between **1.0 and 2.0 ppm**.

(Choice B) Protons adjacent to an electronegative substituent, such as a halogen or oxygen, are deshielded relative to aliphatic protons and generally appear between 2.5 and 4.5 ppm.

(Choices C and D) Vinyl protons (ie, –C=C–H) and protons on an aromatic ring experience an anisotropic effect, causing the protons to be deshielded and have a downfield chemical shift. Vinyl protons do not experience as large of an effect and appear between 4.5 and 6.5 ppm whereas aromatic protons are more downfield between 7.0 and 8.0 ppm.

Educational objective:

The chemical shift of a proton in ^{1}H NMR is given in ppm and depends on the electronic environment around the proton. Protons with an upfield signal (ie, small chemical shift) are more shielded from the magnetic field by their electron cloud. Protons with a downfield signal (ie, larger chemical shift) have structural features that deshield them from the magnetic field.

Time spent:0 sec
Subject:
Organic Chemistry

QID:403495
System:
Separation Techniques, Spectroscopy, and Analytical Methods